

MONROE COUNTY AP, IN, USA

WMO: 724375

Lat: 39.146N

Lon: 86.616W

Elev: 844

StdP: 14.25

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 03893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	9.2	-5.3	4.3	6.0	-0.1	5.6	11.0	23.4	43.5	20.6	44.1	4.0	290	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	91.0	75.1	89.1	74.9	86.8	73.5	78.4	86.9	77.2	85.7	75.7	83.5	7.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	139.5	83.5	74.6	133.4	82.2	73.1	126.7	80.5	42.6	87.0	41.1	85.6	39.7	83.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	17.3	15.6	-5.1	94.1	7.1	4.2	-10.2	97.2	-14.3	99.6	-18.3	102.0	-23.4	105.0		
			-5.8	80.5	7.0	1.7	-10.8	81.7	-14.9	82.6	-18.8	83.6	-23.8	84.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	30.1	33.4	43.6	55.0	64.3	72.2	74.7	73.6	67.2	55.7	44.3	34.5
	DBStd	18.26	12.43	11.75	11.02	9.26	8.02	5.83	5.33	5.00	7.14	8.93	9.76	11.21
	HDD50	2181	625	477	261	52	4	0	0	0	0	42	225	495
	HDD65	5023	1081	885	667	319	115	12	2	3	59	311	622	947
	CDD50	3702	9	12	64	203	446	667	766	732	516	220	54	13
	CDD65	1066	0	0	4	19	93	229	302	270	124	25	0	0
	CDH74	9713	0	0	27	181	796	1968	2835	2486	1211	203	6	0
	CDH80	3258	0	0	2	21	194	658	1045	882	415	41	0	0
Wind	WSAvg	6.5	8.2	7.9	8.0	8.1	6.5	5.6	4.6	4.3	4.6	5.9	7.1	7.5
Precipitation	PrecAvg	44.1	2.8	2.6	3.6	4.1	5.0	4.1	4.6	3.7	3.5	3.0	3.7	3.3
	PrecMax	65.7	9.0	6.1	8.4	10.2	10.7	10.4	11.8	8.6	10.2	8.0	8.8	7.4
	PrecMin	28.4	0.0	0.3	0.7	1.1	0.8	0.6	1.4	1.0	0.2	0.2	0.6	0.4
	PrecStd	8.1	1.9	1.3	2.0	2.0	2.6	2.1	2.4	2.0	2.6	1.9	1.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	68.3	77.5	82.4	88.2	91.3	96.7	94.3	92.0	84.5	73.5	65.9
		MCWB	59.2	59.4	62.3	64.3	72.1	74.6	76.0	74.9	71.4	70.7	61.7	60.2
	2%	DB	59.2	61.6	71.7	79.2	84.6	89.1	91.0	90.4	88.5	79.5	69.5	61.1
		MCWB	53.9	54.6	59.2	63.2	69.2	74.5	76.7	75.2	71.9	65.9	58.8	56.6
	5%	DB	54.4	56.7	66.3	75.4	82.0	86.8	88.4	87.9	84.9	75.0	64.2	55.8
		MCWB	50.7	50.6	55.9	60.9	68.0	73.1	76.3	74.2	71.0	63.7	56.5	51.7
	10%	DB	48.7	52.0	62.1	71.7	78.9	84.0	85.8	84.9	81.3	71.6	60.8	51.1
		MCWB	44.1	45.7	53.4	59.5	66.9	71.7	74.4	72.6	68.4	62.2	53.9	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	60.9	63.7	68.9	74.4	78.5	81.3	79.8	76.8	72.9	64.5	61.2
		MCDB	63.0	66.8	74.7	79.5	83.3	87.6	89.1	88.7	86.6	81.1	69.9	65.1
	2%	WB	55.7	56.3	61.0	65.9	72.3	76.8	79.4	77.9	74.6	69.9	61.1	56.7
		MCDB	58.4	60.2	69.4	74.5	81.4	85.7	87.3	87.0	83.5	76.2	66.4	60.0
	5%	WB	51.0	51.9	58.3	63.6	70.5	75.2	77.8	76.4	72.9	66.5	58.1	52.0
		MCDB	54.0	56.0	65.0	71.2	78.5	83.6	85.7	84.9	81.2	72.4	63.5	55.0
	10%	WB	44.3	45.6	54.7	61.3	68.7	73.7	76.2	74.9	71.0	63.4	54.4	47.0
		MCDB	47.7	50.5	60.7	68.8	76.0	81.3	83.7	82.2	77.6	69.0	59.6	50.2
Mean Daily Temperature Range	5% DB	MDBR	16.1	17.5	19.6	21.4	20.2	19.4	19.5	21.1	22.8	21.3	18.9	14.9
		MCDBR	20.2	22.8	25.0	25.6	23.3	21.7	21.6	23.8	26.6	25.2	24.1	19.0
		MCWBR	17.6	17.6	15.5	13.4	11.0	9.8	9.4	10.9	11.8	13.5	16.5	16.1
	5% WB	MCDBR	19.0	20.4	22.0	19.7	19.3	18.4	18.3	19.7	20.9	20.2	20.6	17.3
		MCWBR	17.7	17.9	14.9	12.6	10.3	9.4	9.3	10.4	10.6	11.9	16.5	16.0
Clear-Sky Solar Irradiance	taub	0.304	0.318	0.349	0.400	0.431	0.442	0.459	0.456	0.421	0.354	0.321	0.297	
	taud	2.464	2.399	2.386	2.248	2.207	2.212	2.163	2.160	2.242	2.440	2.478	2.491	
	Ebn at Noon	277	287	288	278	271	267	261	258	260	270	267	271	
	Edh at Noon	25	31	35	43	45	45	47	46	40	30	25	23	
All-Sky Solar Radiation	RadAvg	579	826	1163	1519	1717	1971	1898	1750	1446	1033	695	489	
	RadStd	62	97	105	109	157	120	115	105	118	105	76	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	N/A

Nomenclature: See separate page